Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 104604
Subject: Electronics Instruments and Measurement
Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The high torque to weight ratio in an analog indicating instrument indicates
- (i) High Friction loss
- (ii) Low friction loss
- (iii) Nothing as regards friction loss
- (iv) None of the above

(b) Which of the following is the most popular method for measuring low resistance?
- (i) ducter ohmmeter method
- (ii) kelvin double bridge method
- (iii) ammeter-voltmeter method
- (iv) potentiometer method

(c) Which of the following method is used for the measurement of Medium Resistance?
- (i) Direct-Deflection method
- (ii) Anderson Bridge
- (iii) Kelvin's double bridge method
- (iv) Carey-Foster bridge method

(d) Which of the following is a balance equation for computing the resistance?
- (i) $R_1 = {}^{R_2}/_{R4}$
- (ii) $R_1 = {}^{R_3}/_{R4}$
- (iii) $R_1 = R_2 R_3$
- (iv) $R_1 = {}^{R_2} R_3/_{R4}$

(e) A Schering bridge can be used for the ___________
- (i) protecting the circuit from temperature rises
- (ii) testing capacitors
- (iii) measuring voltages
- (iv) measuring currents

(f) What is the dependence of frequency on the balance equation?
- (i) varies by a factor of 2
- (ii) depends on the detector used
- (iii) independent
- (iv) depends on the supply magnitude

(g) Which of the following can be measured using Maxwell's Inductance Capacitance Bridge?
- (i) capacitance
- (ii) frequency
- (iii) mutual Induction
- (iv) Induction

(h) The windings of a C.T. are _________
- (i) tied together
- (ii) Shorted
- (iii) wound over one another
- (iv) Grounded

(i) Which of the following device is used to measure power in A.C. circuits?
- (i) ammeter
- (ii) Wattmeter
- (iii) voltmeter
- (iv) ohmmeter

(j) When the moving coil in a Dynamometer type wattmeter deflects _________
- (i) pointer doesn't move
- (ii) current flows
- (iii) voltage is generated
- (iv) pointer moves

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Q.2 (a) What is the need for standards of measurements? How they are classified? Explain [7]

(b) How the unknown frequency is measured using Wien bridge method? [7]

Q.3 (a) State the purpose of calibration of measuring instruments. Explain the procedure of calibration of D.C. Voltmeter by using D.C. Potentiometer. [7]

(b) How do you measure the unknown inductance using Hay's Bridge? [7]

Q.4 (a) How do you measure the unknown inductance using Schering Bridge & also draw the phasor diagram [7]

(b) Draw the possible methods of connecting the pressure coil of a wattmeter and compare the errors. Explain the meaning of 'compensating winding' in a wattmeter and show how they help to reduce the error. [7]

Q.5 (a) List out comparisons between CT's and PT's (Any four). [7]

(b) A current transformer with a bar primary has 300 turns in its secondary winding. The resistance and reactance of the secondary circuit are 1.5Ω and 1.0Ω respectively including the transformer winding with 5 A flowing in the secondary winding, the phase angle error is (assuming magnetizing mmf to be 100 AT and core loss to be 1.5 W) [7]

Q.6 (a) Describe the construction and principle of operation of D' Arsonval type Galvanometer [7]

(b) Write short notes on (a) Kelvin's double bridge (b) Digital frequency meter [7]

Q.7 (a) Describe the measurement of frequency, phase angle and time delay using oscilloscope with suitable diagrams and mathematical expressions. [7]

(b) A moving coil instrument gives a full scale deflection of 10mA, when the potential difference across its terminal is 100mV. Calculate (1) The shunt resistance for a full scale deflection corresponding to 100A (2) The resistance for full scale reading with 1000V. Calculate the power dissipation in each case? [7]

Q.8 (a) Draw the block diagram of an electronic voltmeter and explain its operation [7]

(b) Explain BH curve with a neat diagram. What is Hysteresis loop? [7]

Q.9 (a) Prove that for electrodynamometer type wattmeter true power = $\{\cos\Phi / [\cos\Phi \cos(\Phi - \beta]\}$ x actual wattmeter reading Where $\cos\Phi$ = power factor of the circuit β = tan-1 ($\omega L/R$ where L and R are the inductance and resistance of the pressure coil of the circuit. [7]

(b) Explain with neat sketch; the construction and working principle of Megger. [7]

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